

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Examination – Summer 2024

Course: B. Tech.

Branch: Civil Engineering & Allied

Semester: IV

Subject Code & Name: BTCVC406 / BTCIC406 / BTCEC406 Engineering Geology

Max Marks: 60

Date: 26/06/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Draw a suitable diagram wherever necessary.

Marks

Q.1 Solve Any Two of the following.

- | | |
|---|---|
| A) Write short notes on: - | 6 |
| a. Meandering | |
| b. Tectonic mountains | |
| c. Strato-volcano | |
| B) Write a detailed note on various types of tectonic plate boundaries with neat labelled diagrams. | 6 |
| C) Explain working of river system with various types of drainage patterns. | 6 |

Q.2 Solve Any Two of the following.

- | | |
|--|---|
| A) Write a short note on: - | 6 |
| a. Gneissose texture | |
| b. Batholith | |
| c. Cataclastic metamorphism | |
| B) Write the classification of igneous rocks based on depth, silica and texture. | 6 |
| C) Define sedimentary rock and classification based on grain size and origin. | 6 |

Q.3 Solve Any Two of the following.

- | | |
|--|---|
| A) Write a short note on:- | 6 |
| a. Chevron fold | |
| b. Graben fault | |
| c. Porosity | |
| B) What is the difference between a confined aquifer and unconfined aquifer? | 6 |
| C) Explain various types of joints and civil engineering significance. | 6 |

Q.4 Solve Any Two of the following.

- A) Write a short note on: - 6
 - a. Desk study
 - b. Rock Quality Designation
 - c. Core box design and arrangements
- B) Explain trial pit sample collection method with its advantages and disadvantages. 6
- C) What are the various drilling techniques used for the collection of samples? 6

Q. 5 Solve Any Two of the following.

- A) Write a short note on: - 6
 - a. Tunnel Boring Machine
 - b. Earthen dam
 - c. Immersed tunnel
- B) Explain the various favourable and unfavourable geological conditions for dam site. 6
- C) Explain various types of bridges and their components. 6

***** End *****